Modeling Fisher Participation in Chile's Small Pelagic Fishery

Departamento de Economía
Universidad de Concepción

A multi-species structural approach — FONDECYT project, 2025–2028

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Motivation

- Small pelagic stocks fluctuate sharply — biology, prices, regulation, weather all move year-to-year.
- Standard projection tools assume **constant fishing rates or full quota uptake** — the choice is assumed away.
- But fishers **choose**: whether to participate, which species to target, how many trips to take.
- These choices are **mediated** by quotas (LMCA), portfolio composition, and operating constraints.
- **My approach:** treat fisher participation as a *count process* — estimate it separately by fleet, then propagate to comparative statics.

Today: identification strategy + heterogeneity in revealed responses + roadmap to welfare counterfactuals.

The project — two papers, one structural pipeline

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Paper 1 (today's results) *Reduced-form effort + identified stock dynamics*

- **Negative binomial** trip equation, estimated separately by fleet (LMCA-induced cross-sectional variation in H^{alloc}).
- Bayesian state-space stock dynamics with structural climate shifters (in appendix).
- Comparative statics under counterfactual scenarios.

Target submission 2026.

Paper 2 (today's roadmap) *Welfare counterfactuals*

- Translog **cost** function (intra-annual, by gear).
- **Inverse demand** (IAIDS, 3SLS).
- **Dynamic vessel optimization** $\Rightarrow$ optimal $T_g, h_{g\tau}$ under counterfactual shocks.
- Welfare, prices, **species substitution**.

Estimation 2026–2027.

The two papers share the **same identified stock dynamics**. Paper 1 stops at participation; Paper 2 closes the loop through prices and profits.

Setting and observed heterogeneity

Chile's Small Pelagic Fishery (Centro-Sur)

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Stocks & sectors

- **Anchoveta**, **common sardine**, **jack mackerel**.
- ~94% of national landings; mostly purse seiners.
- Two sectors: **artisanal** (ART), **industrial** (IND).

The LMCA quota regime

- *Límite Máximo de Captura por Armador*: TAC split between sectors with **limited cross-sector transferability**.
- IND quotas partially tradable within sector; ART partly under **RAE**.

Fleet portfolios are very different

- **ART**: ~ {anchoveta, sardine}. Concentrated.
- **IND**: ~ 95% jack mackerel + 5% sardine. Diversified into the *transboundary* stock.

Mobility

- IND: offshore purse-seiners.
- ART: port-anchored, RAE binds spatially.

$\Rightarrow$ Same shock, **different choice sets**.

Observed strategy switching — ART vs IND

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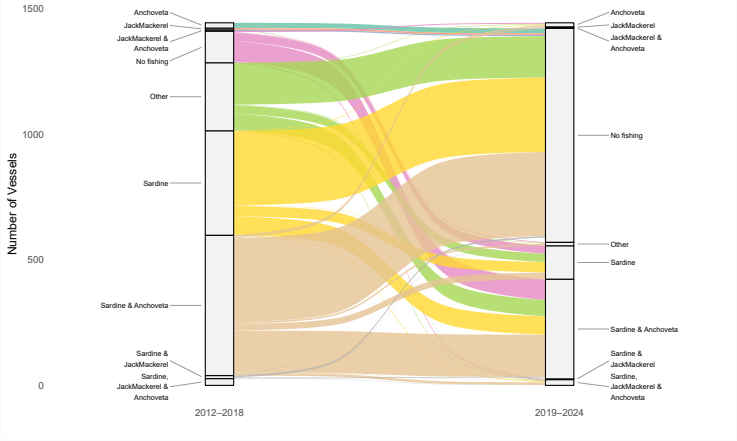
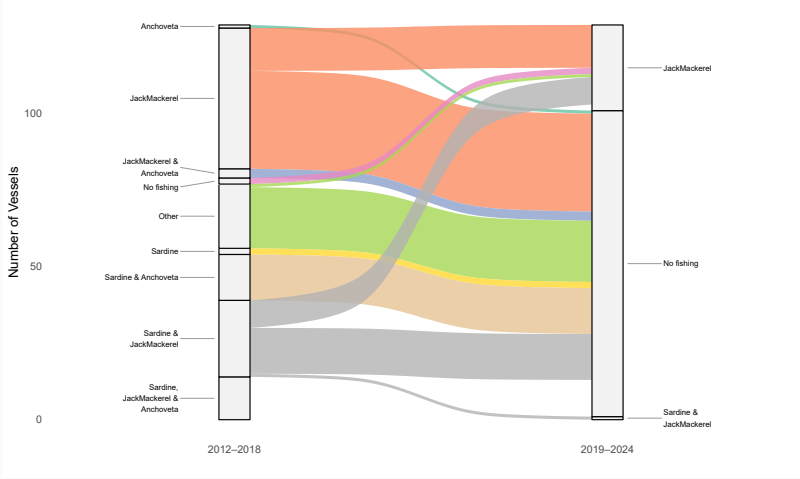
Artisanal



Industrial



Trip-level target-species transitions, IFOP logbooks 2013–2024.

ART: anchoveta↔sardine dominates; jurel essentially absent. IND: rotation across all three. Same fishery, very different revealed choice sets.

Paper 1 — The NB trip equation

A negative binomial model of annual trips

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Specification, by fleet g (ART, IND)

$$T_{vy}^{g} \sim \mathrm{NB}\big(\exp(U_{vy}'\beta_g),\ \theta_g\big)$$

with $U_{vy} = \big[\, p_{sy},\ H_{vy}^{alloc},\ Z_v,\ O_{vy} \,\big]$:

- p_{sy}: ex-vessel prices, by species.
- H_{vy}^{alloc}: vessel-level quota allocation (LMCA $\times$ RAE).
- Z_v: hold capacity, vessel type.
- O_{vy}: bad-weather days, *vedas*.

Why NB, why by fleet?

- Overdispersion σ^2/μ: **22.4** (ART), **4.9** (IND). Poisson rejected at $p < 0.001$.
- Two fleets, two technologies, two choice sets — no pooling.
- Vessel-clustered robust SEs.

Identification of β_g

LMCA shares vary across **vessels** $\times$ **years** (regional TAC moves; vessel share fixed). Reduced-form mapping; full structural identification deferred to Paper 2.

What drives fisher participation? — Estimates by fleet

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Variable	Industrial	Artisanal	Reads as
H_{vy}^{alloc} (allocated quota)	**(+)** ***	**(+)** ***	Quota = binding budget constraint, both fleets.
Price — jack mackerel	**(+)** ***	0	IND oriented to jurel; ART has no exposure.
Price — sardine	0	**(+)** ***	Sardine is ART's primary target.
Price — anchoveta	0	**(–)** ***	Simultaneity sign — addressed in Paper 2.
Hold capacity (log)	0	**(+)** ***	Heterogeneity is *within* the artisanal fleet.
Bad-weather days	0	**(–)** ***	ART vulnerable; IND insulated by vessel size.
Veda days	**(–)** ***	n/a	Largest single coefficient for IND.

Negative binomial, vessel-clustered robust SEs. Sign and significance shown; full table in appendix. $^{*}p < 0.01$.

Two fleets, **completely different participation models**. Same regulatory variables, opposite signs.

Comparative statics — the fleet asymmetry

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Fleet	Counterfactual	$\%\Delta T$ **(marginal)**	$\Pr(\text{portfolio loss} > 50\%)$
Artisanal	mid-century, moderate scenario	**–8.7%**	**0.95**
	end-century, severe scenario	**–9.4%**	**0.99**
Industrial	mid-century, moderate	–0.9%	0.12
	end-century, severe	–0.9%	0.12

- ART: ~9× larger contraction than IND.
- ART: near-certain portfolio loss.
- *Conditional* on no portfolio loss: –2% ART vs –1% IND.

- $\Rightarrow$ Asymmetry runs through **portfolio composition** + **LMCA cross-sector limits**.
- *Not* through trip-level price elasticities.
- IND's protection partly rests on jurel — whose long-run shifter is **n.i.** (deferred).

Paper 2 — closing the loop

Three structural extensions ⇒ welfare counterfactuals

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(i) Trip-level cost (translog)

$$C_{vg} = \sum_i \alpha_{g,X_i} X_{ivg} + \tfrac{1}{2} \sum_{i,j} \alpha_{g,X_i X_j} X_{ivg} X_{jvg}$$

with $X_{vg} = [w; h_{vg}; x; Z_v; Env]$. SUR per gear.

(ii) Inverse demand (IAIDS)

$$\ln p_{iy} = \sum_j \gamma_j \ln h_{j,y} + \gamma_H \ln H_y + \gamma_{FM} \ln P_y^{FM} + \epsilon_{iy}$$

3SLS; $h_{j,y}$ instrumented by SST, CHL, fuel prices.

⇒ resolves the anchoveta-price puzzle from Paper 1.

(iii) Vessel dynamic optimization

$$\max_{h_{g\tau}, T_g} \sum_{\tau=1}^{T_g} \delta^\tau \{P(h)\, h_{g\tau} - C_g(h_{g\tau} \,|\, w, x, Z, Env)\}$$

$$\text{s.t.} \quad q_{t+1} = \omega\, \bar{q} - \sum h_{g\tau} \geq 0$$

Counterfactuals it enables

- **Equilibrium price adjustment** under reduced supply.
- **Species substitution** within the harvest decision.
- **Policy levers**: LMCA cross-sector reform, RAE redesign, state-contingent TACs — each evaluated as an *adaptation instrument*, not just allocation.

Takeaways

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- **Paper 1**: a fleet-specific NB model of annual trips reveals **two completely different participation technologies** in the same fishery — ART and IND respond to different prices, different operating constraints, different scales.
- **Paper 1**: under counterfactual shocks, the ART:IND participation asymmetry is **~9:1**. The mechanism is **portfolio composition** × **LMCA cross-sector limits**, not differential elasticities.
- **Paper 2**: closes the loop — cost, demand, dynamic optimization — to deliver welfare counterfactuals and evaluate policy reforms (LMCA transferability, RAE redesign).
- **Joint message for S9**: adaptive capacity is not a property of fishers; it is a property of the **bundle (portfolio × permit × institution × market)** they sit in. The empirical model has to respect that structure.

¡Gracias! / Thank you

Questions?

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Appendix

Appendix A · Stock dynamics — spec & priors

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Process (Schaefer with shifter on r):

$$B_{i,t+1} = B_{i,t} + r_{i,t} B_{i,t}\left(1 - \frac{B_{i,t}}{K_i}\right) - C_{i,t} + \varepsilon_{i,t}$$

with structural climate shifter

$$r_{i,t} = r_i^0 \exp\left(\rho_i^{SST} \Delta SST_{t-1} + \rho_i^{CHL} \Delta \log CHL_{t-1}\right)$$

Observation (log-normal): $\log B_{i,t}^{obs} = \log B_{i,t} + u_{i,t}$.

Three nested specs: **ind**, **omega**, **full**.

Priors:

- $r_i^0, K_i, \sigma_{proc}, \sigma_{obs}$: from IFOP / SPRFMO single-species assessments.
- $\rho_i^{SST}, \rho_i^{CHL} \sim \mathcal{N}(\hat{\rho}_i^{stress}, 1)$.
- Ω: LKJ.

Estimation: Stan / HMC, 4 chains $\times$ 2000 post-warmup; $\hat{R} < 1.01$, $ESS > 400$.

Appendix B · Climate shifters — posterior identification

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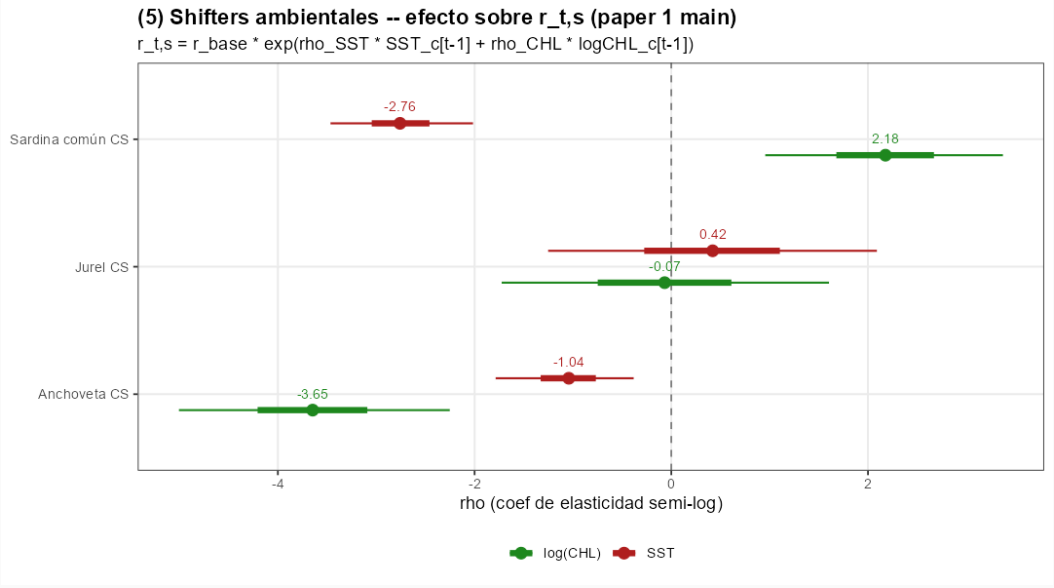

Posterior densities of ρ^{SST} and ρ^{CHL}, full multi-species specification. Anchoveta and common sardine: priors dominated by data. Jack mackerel: posteriors track priors $\rightarrow$ **n.i.** and reported as such throughout.

Appendix C · Stock productivity comparative statics

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Stock	Scenario	$\Delta r/r_0$ (median)	90% CI	$\Pr(\Delta < 0)$
Anchoveta	SSP2-4.5, mid-century	**–51%**	[–74%, –17%]	0.99
	SSP5-8.5, end-century	**–89%**	[–98%, –50%]	0.99
Sardine	SSP2-4.5, mid-century	**–90%**	[–95%, –82%]	1.00
	SSP5-8.5, end-century	**–100%**	[–100%, –99%]	1.00
Jack mackerel	all scenarios	n.i.	n.i.	n.i.

Appendix D · Why is jack mackerel n.i.?

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- Posteriors track priors across alternative spatial windows and biomass series.
- The CS biomass signal carries little information about climate shifters — pre-2002 acoustic gap, censored 2012/2015 observations, transboundary SPRFMO-scale dynamics.
- Reduced-form linear stress tests on growth increments do **not** close the hindcast under SST/CHL forcing alone, consistent with structural non-identification.
- **Implication:** climate-driven projection for jurel CS must come from a **transboundary, game-theoretic** model (cf. Session S9, bullet 2), not a CS-only structural fit.

Appendix E · Model comparison (LOO)

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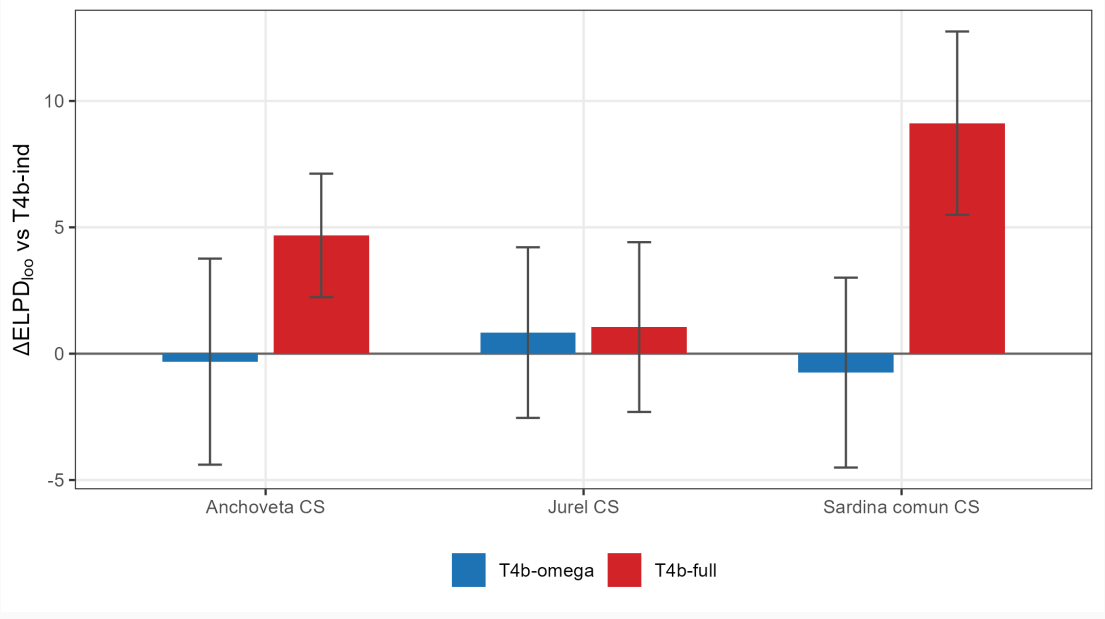
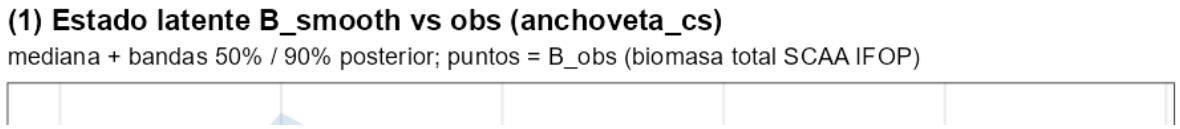
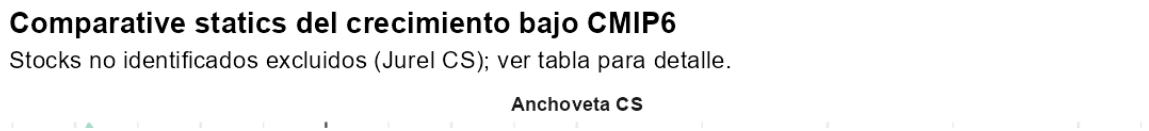


LOO ELPD across specs: **ind** (Ω diagonal, $\rho = 0$), **omega** (Ω full, $\rho = 0$), **full** (Ω full, $\rho \neq 0$). Full wins at $\Delta\mathrm{ELPD} \approx 14$–$18$; LFO one-step-ahead essentially tied.

Appendix F · Posterior predictive check



Appendix G · CMIP6 ensemble — ridgeline



Appendix H · Paper 2 — data inputs

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Already in hand (Paper 1):

- Trip-level catch and price data (IFOP, 2013–2024).
- Vessel characteristics & ports of operation.
- Environmental covariates (Copernicus, GlobColour).

To request / acquire for Paper 2:

- Variable-input prices: crew wage, diesel cost, ice & supplies.
- Quota prices (auction or secondary markets, if available; informs forward-looking behavior cf. Birkenbach et al. 2024).
- Permits by vessel; ITQ holdings by sector.
- LMCA quota allocations and inter-fleet transfers (SERNAPESCA).
- Fish-meal world price — demand instrument.